

Cosmic Impulse



The game, explained without code

“Build the spaceship you’ve always wanted. Design your engines, weapons, armor, power systems, interiors, and almost anything else you can imagine. Then take your creation into a massive, to-scale solar system to explore, mine, build, and battle other players in their own spaceships.”

From the Vision

Version: first playable foundation (rule set orbital-arena-0.1, fingerprint 163fa166). Generated October 07, 2026. Free and open source under the GNU General Public License, version 2.

1. What Cosmic Impulse is

Cosmic Impulse is a spaceship game where the physics is real and the creativity is yours. You design a ship, decide how it looks, works and feels, and fly it in a solar system built to scale. Everyone plays by the same rules of physics, so the only way to get stronger is the way real engineers do it: collect energy and resources, and make smarter tradeoffs.

Three ideas from the Vision shape every decision:

- **Make it yours.** Total freedom over your ship's shape, layout, cockpit, displays, controls, interiors, sounds and interface. You can build your own targeting screens, warning systems and automation from the data your ship's sensors provide.
- **Same rules.** Customization never changes what a ship can physically do. Speed, mass, energy, weapon output and sensor range all follow one shared rule set that is identical for every player.
- **Almost unlimited creation.** Instead of only unlocking a fixed list of recipes, players should eventually be able to invent their own components, processes and technologies, still bound by the same physics.

The one-sentence version: energy is the currency, physics is the referee, and your imagination decides what to build with them.

Free and open, for real

Cosmic Impulse is 100% free to play. Real money cannot buy ships, weapons, resources, upgrades or any competitive advantage. The code is open source under the GNU GPLv2, so anyone can read it, improve it, and host their own version.

2. The same rules for everyone

The rules define the world, its physics, its resources, and how technology scales. They are fixed, public, and the same for everyone:

- **A to-scale solar system** with the Sun at its centre.
- **Real movement:** Newton's laws and orbital mechanics at everyday speeds, with special relativity and time dilation taking over as ships approach the speed of light. Practical approximations are used where needed for performance and fun.
- **A fixed Sun.** It always produces the same energy; the closer you go, the more of it each solar panel catches.
- **300× server speed**, so collecting energy, processing resources and travelling between worlds happen much faster than in real life.
- **Resource extraction** from planets, asteroids and other bodies.
- **Defined scaling** for engines, weapons, armour, power, storage and every other component, with more flexibility as technology advances.

How the game keeps everyone honest

All rules live in one place and are published as the **Rules file**. The game turns that file into a short code called a **fingerprint**, shown on the main menu. Two games can only fight each other if their fingerprints match. Anyone may change the code, but a changed rule set simply cannot join a battle with the official one.

Players never type in their ship's statistics. A design is a recipe (a ship class and a few choices), and every computer recalculates what that recipe can do using the shared physics. A design cannot claim to be stronger than its physics allows.

3. Freedom in layers: easy to start, no ceiling

Cosmic Impulse should be fun for someone who knows no physics, and endlessly deep for someone who loves it. The answer is four layers of design. Every layer produces a real design that runs on the same physics, so nobody is playing a "simplified" game.

Layer	What you do	Physics you need
1. Pick	Choose a ready-made ship class: Scout, Brawler, Sniper or Missile carrier.	None

Layer	What you do	Physics you need
2. Tune	Move plain-language sliders such as Punch or Endurance, Armoured or Agile.	None
3. Modify	Swap and edit individual components, with every number explained.	A little, learned in the game
4. Invent	Design new engines, processes, recipes and automation from basic physical building blocks.	As much as you like

Results are always explained in game terms, not equations. For example, straight from the game:

- **Brawler:** 0.58 g with full tanks, 0.94 g when they run dry.
- **Brawler:** Turns around in 5.3 s.
- **Sniper:** Can fire every laser for 12 min before overheating.

A shared blueprint library lets beginners fly brilliant designs made by experts, so knowing physics is a way to invent, never a requirement to compete.

Path 1 or Path 2?

The Vision leaves open whether players should design their own systems from scratch (Path 1) or combine a massive library of detailed components (Path 2). The layered approach offers a way to have both:

- Layers 1 to 3 are **Path 2**: a large, growing library of components and blueprints to pick, tune and combine.
- Layer 4 is **Path 1**: the tools to invent new components and processes from physical building blocks.
- Both sit on one physics engine, so anything invented in Layer 4 can join the library for everyone else.

This is a recommendation, not a decision. The choice between the paths remains with the project owner.

No free lunch

Every choice trades one thing for another. In the current version, each slider is backed by real physics:

Slider	You gain	You give up
Endurance / Punch	Thrust (oxygen burned in the nuclear exhaust)	Total speed change, because the exhaust is slower

Slider	You gain	You give up
Agile / Armoured	Time before your armour burns through	Acceleration, because armour is heavy
Steady / Powerful	Faster weapon recharge	Mass, and some extra heat
Small target / Big radiators	Faster cooling	Mass, and large, fragile targets
Light / Long legs	More total speed change	Acceleration, because propellant is heavy
Short fights / Long fights	Longer firing before overheating	Mass, because water is heavy

4. Ship battles

Space combat in Cosmic Impulse is realistic and readable. Battles are decided by distance, heat and timing, not hit points.

Heat is the heart of every fight

Engines, lasers and railguns all turn part of their energy into waste heat. In space there is no air to carry it away: a ship can only shed heat by glowing it off its radiators. Big radiators cool you quickly, but they are large, fragile targets. Folding them away protects them, but then you heat up. A tank of water boils to soak up heat in a pinch, and a ship that gets too hot loses its weapons, then its crew.

Every battle asks the same question in a hundred ways: **how much heat can I afford right now?**

The weapons

Weapon	How it works	Strength and weakness
Laser	Light hits instantly. Its reach depends on mirror size, because a bigger mirror keeps the beam tight over a longer distance.	Cannot be dodged, but makes a lot of heat and must burn through armour.
Railgun	Fires metal slugs at 6 km/s that punch through armour.	Devastating up close; at long range the target simply moves out of the way.

Weapon	How it works	Strength and weakness
Missile	A guided rocket that chases its target and hits at kilometres per second.	One hit can gut a ship, but lasers can shoot missiles down.
Point defence	Your lasers automatically fire at incoming missiles.	Spends energy and heat that could otherwise go into attacking.

Armour is tracked separately on each side of the ship, so turning a fresh face toward the enemy matters. Once armour is breached, hits damage what is inside: the crew, the reactor, the engine, the generator, the heat sink and the weapons.

Every ship has a counter

Nobody designed a rock-paper-scissors table. It emerges from the physics. These are the actual results of the game's AI pilots fighting each other with the default designs, starting 400 km apart:

Matchup	Winner	How it ended
Scout vs Brawler	Brawler	after 14 min: the reactor was destroyed
Scout vs Sniper	Sniper	after 8 min: the reactor was destroyed
Scout vs Missile carrier	Scout	after 20 min: lost on points after 20 minutes
Brawler vs Sniper	Brawler	after 15 min: the reactor was destroyed
Brawler vs Missile carrier	Missile carrier	after 20 min: lost on points after 20 minutes
Sniper vs Missile carrier	Sniper	after 15 min: the reactor was destroyed

Every class wins at least one matchup and loses at least one. Brawlers survive the approach and win up close; snipers destroy anything they can keep at range; missile carriers overwhelm ships with weak point defence; scouts carry enough laser to shred missile salvos.

5. What you can play today

This first version is the **Arena**: pick a nuclear-age warship, tune it, and fight the AI or a friend. It is built on the same physics engine the full game will use.

Scout

Fast and light. Its laser shreds missiles and thin armour.

Acceleration	0.31 g with full tanks, 0.67 g when they run dry.
Fuel	5.5 km/s of total speed change: about 21 min of full thrust.
Turning	Turns around in 5.0 s.
Laser	14 MW beam at full strength out to 290 km; a quarter as strong at twice that.
Heat	Can fire every laser for 30 min before overheating.
Engine heat	Can burn the engine for 3.8 h before overheating.
Armour	Each armour patch survives 49 s under a 50 MW laser.
Mass	194 t fuelled, 90 t empty.

Brawler

Heavy armour and twin railguns. Wins up close, if it survives the approach.

Acceleration	0.58 g with full tanks, 0.94 g when they run dry.
Fuel	3.3 km/s of total speed change: about 8 min of full thrust.
Turning	Turns around in 5.3 s.
Laser	3.5 MW beam at full strength out to 193 km; a quarter as strong at twice that.
Railguns	2 × 18 MJ slugs that punch through 900 kg/m ² of armour. Hard to land beyond ~20 km against a ship that dodges.
Heat	Can fire every laser for 5.7 h before overheating.
Engine heat	Can burn the engine for 50 min before overheating.
Armour	Each armour patch survives 144 s under a 50 MW laser.
Mass	236 t fuelled, 146 t empty.

Sniper

A huge laser behind a 4-metre mirror. Deadly at long range, fragile up close.

Acceleration	0.21 g with full tanks, 0.34 g when they run dry.
Fuel	4.1 km/s of total speed change: about 26 min of full thrust.
Turning	Turns around in 5.3 s.
Laser	53 MW beam at full strength out to 773 km; a quarter as strong at twice that.
Heat	Can fire every laser for 12 min before overheating.
Engine heat	Can burn the engine for 13.7 h before overheating.
Armour	Each armour patch survives 65 s under a 50 MW laser.
Mass	233 t fuelled, 143 t empty.

Missile carrier

Salvos of kinetic missiles, guarded by point-defence lasers.

Acceleration	0.33 g with full tanks, 0.61 g when they run dry.
Fuel	4.8 km/s of total speed change: about 18 min of full thrust.
Turning	Turns around in 5.2 s.
Laser	5.3 MW beam at full strength out to 193 km; a quarter as strong at twice that.
Missiles	24 kinetic missiles (6 km/s each). One hit can gut a ship, but lasers can shoot them down.
Heat	Can fire every laser for 2.4 h before overheating.
Engine heat	Can burn the engine for 1.6 h before overheating.
Armour	Each armour patch survives 97 s under a 50 MW laser.
Mass	226 t fuelled, 122 t empty.

Controls in a sentence

You steer relative to your enemy (toward, away, circle, climb, dive, brake) and your ship turns its engine to match. Lasers and railguns fire while you hold their keys, missiles launch in salvos, and two switches decide your heat strategy: point defence on or off, and radiators extended or folded away. A help panel in the battle lists every key.

Playing with a friend

One player hosts and shares a short room code; the other joins with it. If a network blocks that, the two players can exchange two text codes by chat or email instead, with no server at all. Before the battle starts, both games compare rule fingerprints and recompute each other's ship from its design. During the battle, each player's computer decides what happens to their own ship, and any reported hit is capped at what the attacker's weapons could physically deliver.

Time in battle

The world runs at 300× real time, which is perfect for travel and industry but would make a ten-minute dogfight last two seconds. The Arena therefore fights in real time. Making this an official rule (ships in combat with each other drop to real time) is a proposal awaiting the project owner's decision.

6. Where it goes next

The technology journey

Players begin in a **Solar Punk** era and progress through **Ion**, **Nuclear**, **Antimatter** and **Beyond**. Technology is discovered through experimentation, research, resource gathering and invention, not a fixed tree. Research improves how close your machines come to physical limits (a better solar cell, a hotter radiator, a lighter reactor), but nothing ever passes those limits. The Arena's ships belong to the Nuclear era.

Roadmap

Step	What it adds
Physics kernel (done)	Engines, power, heat, armour and weapons computed from one shared rule set.
Arena (done)	Layer 1 and 2 ship design, AI opponents, two-player online battles.
Deeper design	Layer 3 component editing, then Layer 4 invention from physical building blocks.
The solar system on the kernel	The existing sandbox prototype (orbits, mining, solar energy, research) rebuilt on the same physics.
A shared world	A server that runs the same physics for everyone and keeps the official record of energy and resources.

Step	What it adds
Your own interface	A Home-Assistant-style system of sensors, controls, dashboards and automation built by players.
Technology by discovery	Research as experimentation across the Solar Punk to Beyond eras.

Decisions for the project owner

- **Path 1 or Path 2** for creation (Section 3 suggests combining them in layers).
- **Combat time:** should fights between ships always drop to real time, while the rest of the world runs at 300×?
- **Player-versus-player rules** in the shared world: safe zones near Earth, protection for offline players, or consent to fight.
- **Arena time limit:** currently 20 minutes, after which the ship that dealt more damage wins.

7. A few words, explained

Word	Meaning
Delta-v (total speed change)	How much a ship can change its velocity before its fuel runs out. The true measure of how far it can go.
Exhaust speed	How fast an engine throws out propellant. Faster means more speed change from the same fuel, but less push per unit of power.
Radiator	A panel that glows heat away into space. The only way a spaceship can cool down.
Heat sink	Something that soaks up heat for a while, such as water that boils away.
Point defence	Lasers that automatically shoot down incoming missiles.
Blueprint	A shareable ship design: the choices you made, from which anyone can recompute what the ship can do.
Rules fingerprint	A short code that identifies the exact rule set. Games only fight if theirs match.

“The goal is not simply to give players a large selection of things to build, but to eventually give them the tools to invent what comes next.” (the Vision)